

Experiment 2: Develop and compare CLI, GUI, and Voice User Interfaces (VUI)

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Aim

To design and develop applications using GUI, CUI, and VUI interfaces for understanding different user interaction methods.

Objective

- To create a GUI-based To-Do List application.
- To create a CUI-based To-Do List application.
- To develop a VUI-based Calculator using voice commands.
- To study the usability of different user interfaces.

1. GUI To-Do

List Description

The GUI To-Do List provides a graphical interface where users can add and remove tasks using buttons and text fields.

Technologies Used

- Python
- Tkinter

GUI To-Do List Code

```
from tkinter import *

# Create window
root = Tk()
root.title("GUI To-Do
List")
root.geometry("300x300")

# Function to add task
def add_task():
    task = entry.get()
```

```

if task != "":
    listbox.insert(END, task)
    entry.delete(0, END)

# Function to remove task
def remove_task():
    selected = listbox.curselection()
    listbox.delete(selected)

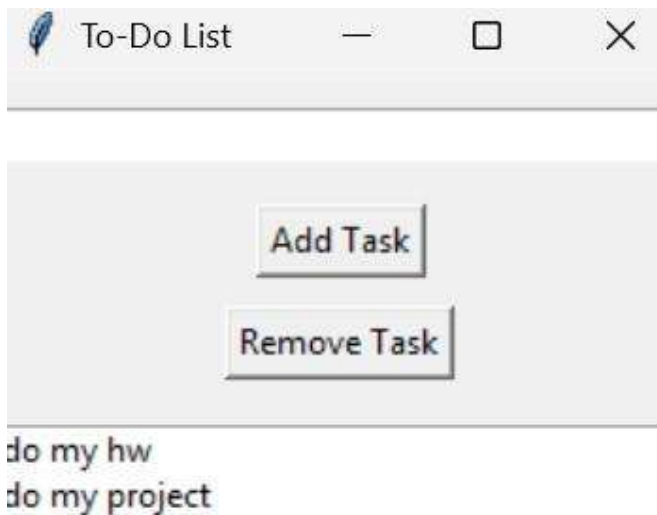
# Entry field
entry = Entry(root, width=30)
entry.pack(pady=10)

# Buttons
Button(root, text="Add Task", command=add_task).pack(pady=5)
Button(root, text="Remove Task", command=remove_task).pack(pady=5)

# Task list
listbox = Listbox(root, width=35, height=10)
listbox.pack(pady=10)

root.mainloop()

```



2. CUI To-Do

List Description

The CUI To-Do List allows users to manage tasks through terminal commands.

Technologies Used

- Python

CUI To-Do List Code

```
tasks = []
```

```
while True:
```

```
    print("\nTo-Do List")
    print("1. Add Task")
    print("2. Remove Task")
    print("3. View Tasks")
    print("4. Exit")
    choice = input("Enter your choice: ")
```

```
    if choice == "1":
```

```
        task = input("Enter task: ")
        tasks.append(task)
        print("Task added")
```

```
    elif choice == "2":
```

```
        task = input("Enter task to remove: ")
        if task in tasks:
            tasks.remove(task)
            print("Task removed")
        else:
            print("Task not found")
```

```
    elif choice == "3":
```

```
        print("\nTasks:")
        for t in tasks:
            print("-", t)
```

```
    elif choice == "4":
```

```
        print("Exiting...")
        break
```

```
    else:
```

```
        print("Invalid choice")
```

```
C:\Users\Lenovo\Downloads>notepad tt.py
C:\Users\Lenovo\Downloads>python tt.py

Options:
1. Add Task
2. View Tasks
3. Remove Task
4. Exit
Enter your choice: 1
Enter task: do my hw
Task 'do my hw' added.

Options:
1. Add Task
2. View Tasks
3. Remove Task
4. Exit
Enter your choice: 2

Your tasks:
1. do my hw
```

3. VUI

Calculator

Description

The VUI calculator performs arithmetic operations using voice commands and provides voice output results.

Technologies Used

- Python
- SpeechRecognition
- pyttsx3

VUI Calculator Code

```
import speech_recognition as sr
import pyttsx3

# Initialize recognizer and speaker
recognizer = sr.Recognizer()
engine = pyttsx3.init()

# Speak function
```

```

def speak(text):
    engine.say(text)
    engine.runAndWait(
    )

# Take voice input
def take_command():

    with sr.Microphone() as source:
        print("Speak your calculation")
        audio =
        recognizer.listen(source)

    try:
        command = recognizer.recognize_google(audio)
        print("You said:", command)
        return command.lower()

    except:
        speak("Could not understand")
        return ""

# Calculator function
def
calculate(command):

    try:

        if "plus" in command:
            numbers = command.split("plus")
            result = int(numbers[0]) + int(numbers[1])

        elif "minus" in command:
            numbers = command.split("minus")
            result = int(numbers[0]) - int(numbers[1])

        elif "multiply" in command:
            numbers = command.split("multiply")
            result = int(numbers[0]) * int(numbers[1])

        elif "divide" in command:
            numbers = command.split("divide")
            result = int(numbers[0]) /
            int(numbers[1])

        else:
            speak("Operation not supported")
            return

    print("Result =", result)

```

```
        speak(f"The result is {result}")

    except:
        speak("Calculation error")

# Main
command = take_command()
calculate(command)
```

```
Assistant: Say add, subtract, multiply, or divide
--- Listening (Attempt 1) ---
User said: divide
Assistant: What is the first number?
--- Listening (Attempt 1) ---
User said: 11
Assistant: What is the second number?
--- Listening (Attempt 1) ---
Assistant: I didn't catch that, let's try again.
Assistant: What is the second number?
--- Listening (Attempt 2) ---
User said: three three
Assistant: That wasn't a valid number.
Assistant: What is the second number?
--- Listening (Attempt 1) ---
Assistant: I didn't catch that, let's try again.
Assistant: What is the second number?
--- Listening (Attempt 2) ---
User said: zero zero
Assistant: That wasn't a valid number.
Assistant: What is the second number?
--- Listening (Attempt 1) ---
Assistant: I didn't catch that, let's try again.
Assistant: What is the second number?
--- Listening (Attempt 2) ---
User said: four
Assistant: The result is 2.75
Final Answer: 2.75
```

Working Process

GUI To-Do List

1. User enters a task.
2. Clicking Add Task stores the task.
3. Remove Task deletes selected tasks.

CUI To-Do List

1. User enters choices through terminal.
2. Tasks can be added, viewed, or removed.

VUI Calculator

1. User speaks the calculation.
2. System recognizes voice input.
3. Result is calculated and spoken back.

Result

The GUI and CUI To-Do List applications successfully managed tasks, while the VUI calculator successfully performed arithmetic operations using voice commands.